

High Consequence Event Management

TOPIC LEARNING OBJECTIVES

Upon successful completion of this topic, the student will be able to:

1. Discuss the relationship between effectively managing High Consequence Events (HCE) and the core tenets of the Navy Leader Development Framework.
2. Compare and contrast the Fukashima nuclear disaster with similar types of high consequence activities conducted by the Navy.
3. Discuss the role organizational culture plays in contributing to or reducing the occurrence of high consequence events.
4. Understand Heinrich's Triangle as it relates to accidents and near-misses and the role the human element plays in causing or preventing them from happening.
5. Understand The HCEPF Principles and Values.
6. Recognize the different types of Human Element Weaknesses (and their corresponding positive attribute) and discuss how they can contribute (or prevent) to the occurrence of a high consequence event.
7. Explain what it means to be a Risk Aware organization.
8. Review and discuss several real-world HCE case studies and the associated human element weaknesses.
9. Discuss tools available to use the HCEPF within your organizations and teams to create risk aware organizations.

STUDENT PREPARATION

Student Support Material

1. Strengthening SSP: The High Consequence Events Prevention Framework, February 2023 [focus on pp. 1 – 52]
2. HCEPF Tools Reference Guide Version 5s, Feb 2025 [read all front matter and pick 2-3 tools to read in detail]
3. HCEPF Fitness Regime Version 3, May 2019 [skim]

Primary References

1. Strategic Systems Program HCE Case Study
2. "Driving Toward "0"; Best Practices in Corporate Safety and Health"; The Conference Board, 2003
3. https://www.retroreport.org/video/major-malfunction-lessons-from-challenger/

Additional References

1. DOE Human Performance Improvement Handbook, VOL II
2. How Habits Work https://youtu.be/W1eYrhGeffc
3. Using Habit Insights in Military Strategy https://youtu.be/s7waGhujn44

Fukushima (March 2011): Where Human Error Prevailed

- Natural disaster followed by man-made disaster
  - ***Shutdown*** an industry key to Japanese economic security
- Production–focused approach to risk by the utility
- A dichotomy:
  - Population safety vs Reactor safety
- Why did the cores melt?



Fukushima Daiichi after 9.1M_w earthquake and 15m tsunami caused nuclear reactor accidents

*"The fundamental causes are to be found in the ingrained conventions of Japanese culture: our **reflexive obedience**; our **reluctance to question authority**; our devotion to '**sticking with the program**'; our **groupism**; and our **insularity**"*

Kiyoshi Kurokawa, Chairman, Fukushima Independent Investigation Commission

Why High Consequence Events Prevention?

- If a technically strong organization can identify the cultural flaws that have arisen within its operations, and
- If that organization and its people (from junior team members to top management) commit themselves to engaging those flaws and to building a stronger culture founded in basic principles of organizational success…

→ ***<u>The organization can preempt many mistakes that otherwise could threaten its success</u>***

Key to success: a strong culture and a strong technical foundation

Developing the Culture



What's the right approach?

"There are no organizations without institutional habits. There are only places where they are ***deliberately designed***, and places where they are created without forethought."

Charles Duhigg, *The Power of Habit: Why We Do What We Do in Life and Business*

The "deliberately designed" option is necessary to shape a useful organizational culture

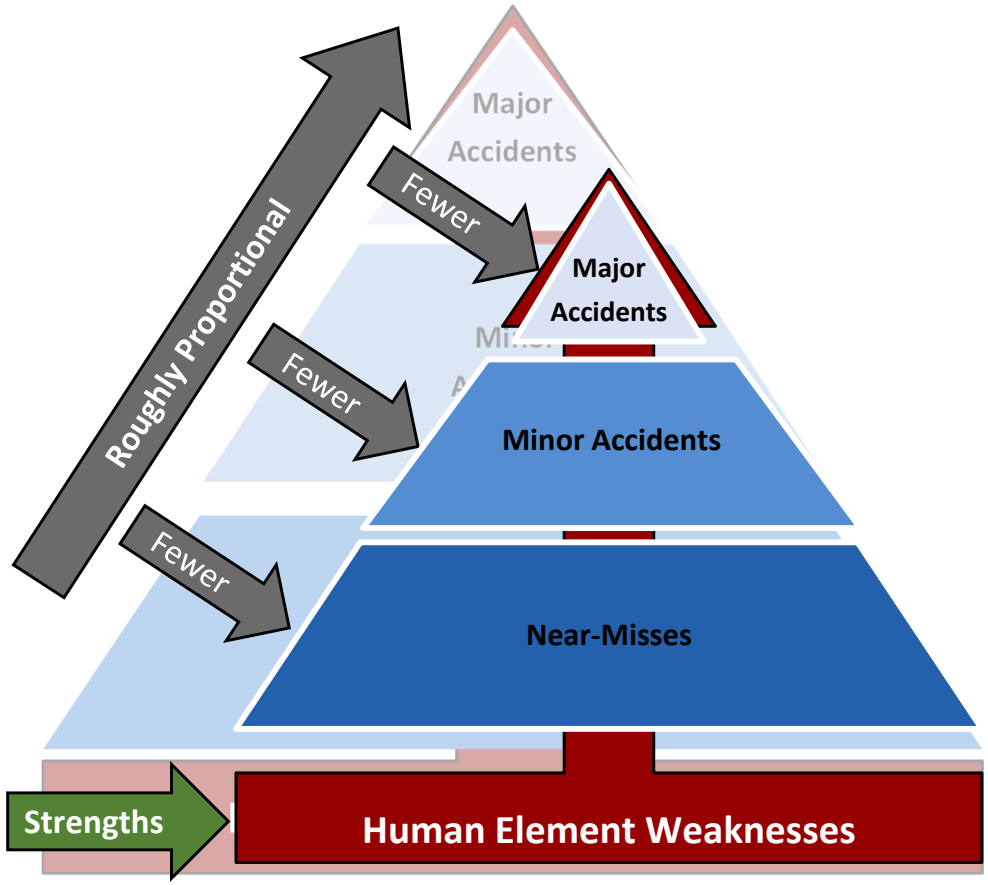
Risk Aware Culture



A deliberately designed culture that fosters day-to-day use of Risk Aware Behaviors to ***make better decisions and reduce human error***, one in which ***all members feel responsible*** for promoting those behaviors and eliminating the corresponding Human Element Weaknesses. A culture in which people are ready to ***learn from past mistakes***, eager to identify and understand the risks to their mission, and ***empowered to act*** to mitigate those risks

Heinrich's Triangle

Apex Event: **FUKUSHIMA**

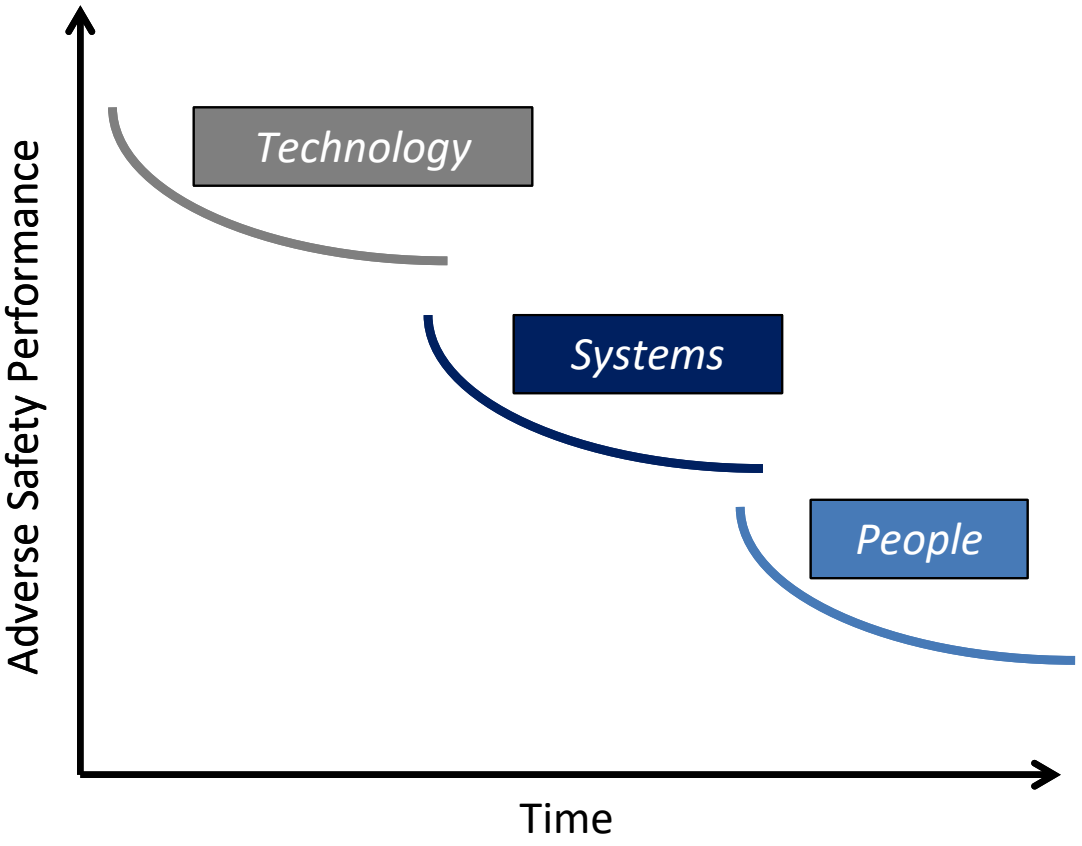


- ***Almost all accidents are sourced in some aspect of human error***
- Proportionality typical between major accidents, minor accidents, and near-misses
- The ultimate disaster is at the apex
- The same causal factors are behind incidents at all levels (***Weaknesses in the Human Element***)
- Learn from events at the lowest level and develop deliberately designed habits:
  - Naturally reduce likelihood of major events
  - Move away from the apex

Prevent apex events by tracking and correcting small failures

Three Levels of Performance Focus

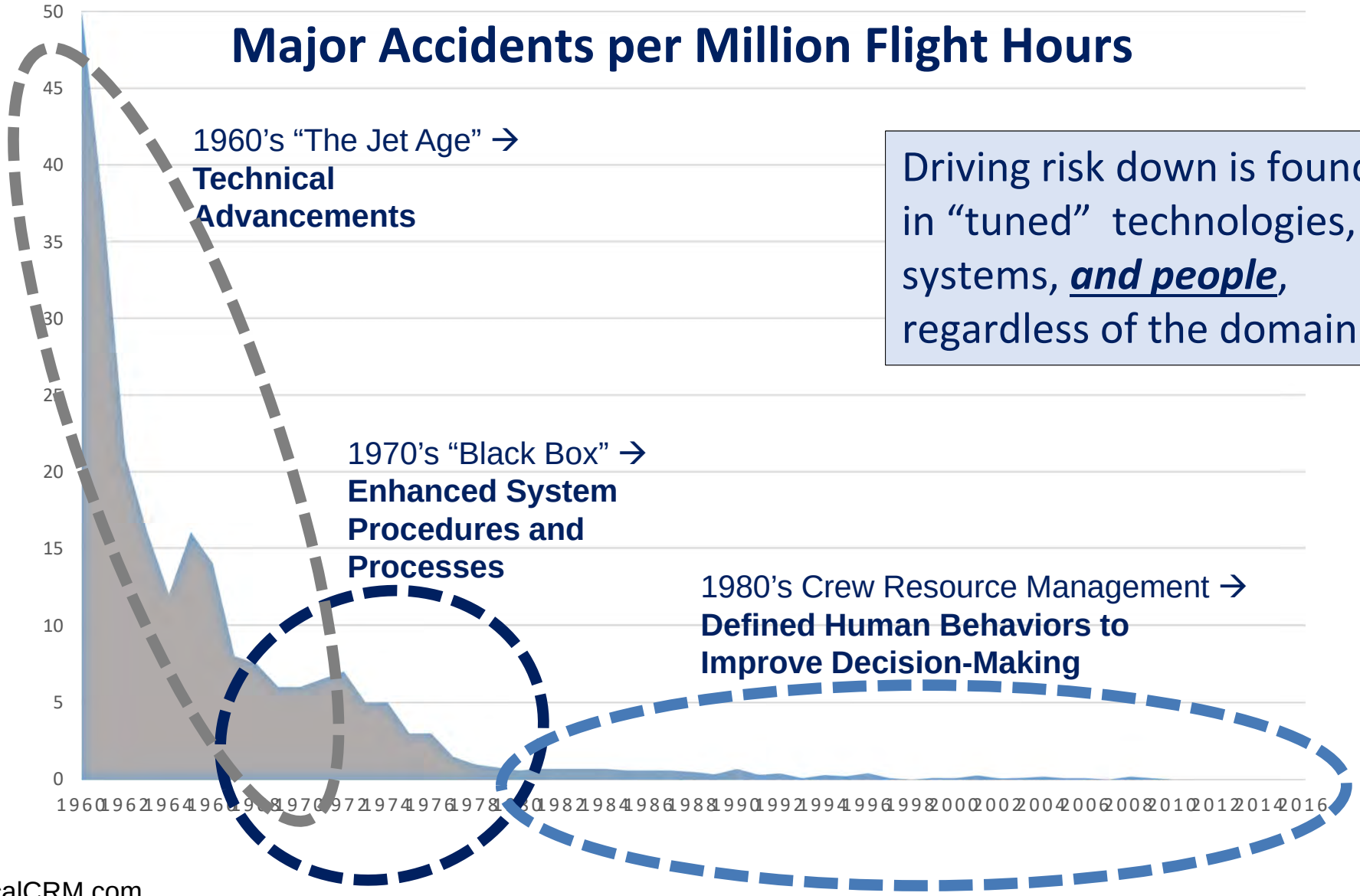
Experience at ExxonMobil



- ***Technology focus***: safety at the component level
- ***Systems focus*****:** safety through system of systems integration
  - Procedures
  - Processes
  - Compliance
- ***People focus***: safety through personnel behavior. Factors:
  - Leader commitment
  - Knowledgeable and skilled application of tools
  - Workforce buy-in
  - Personal accountability and willingness to intervene

All three domains are important – a holistic approach to risk and mission assurance requires attention to each

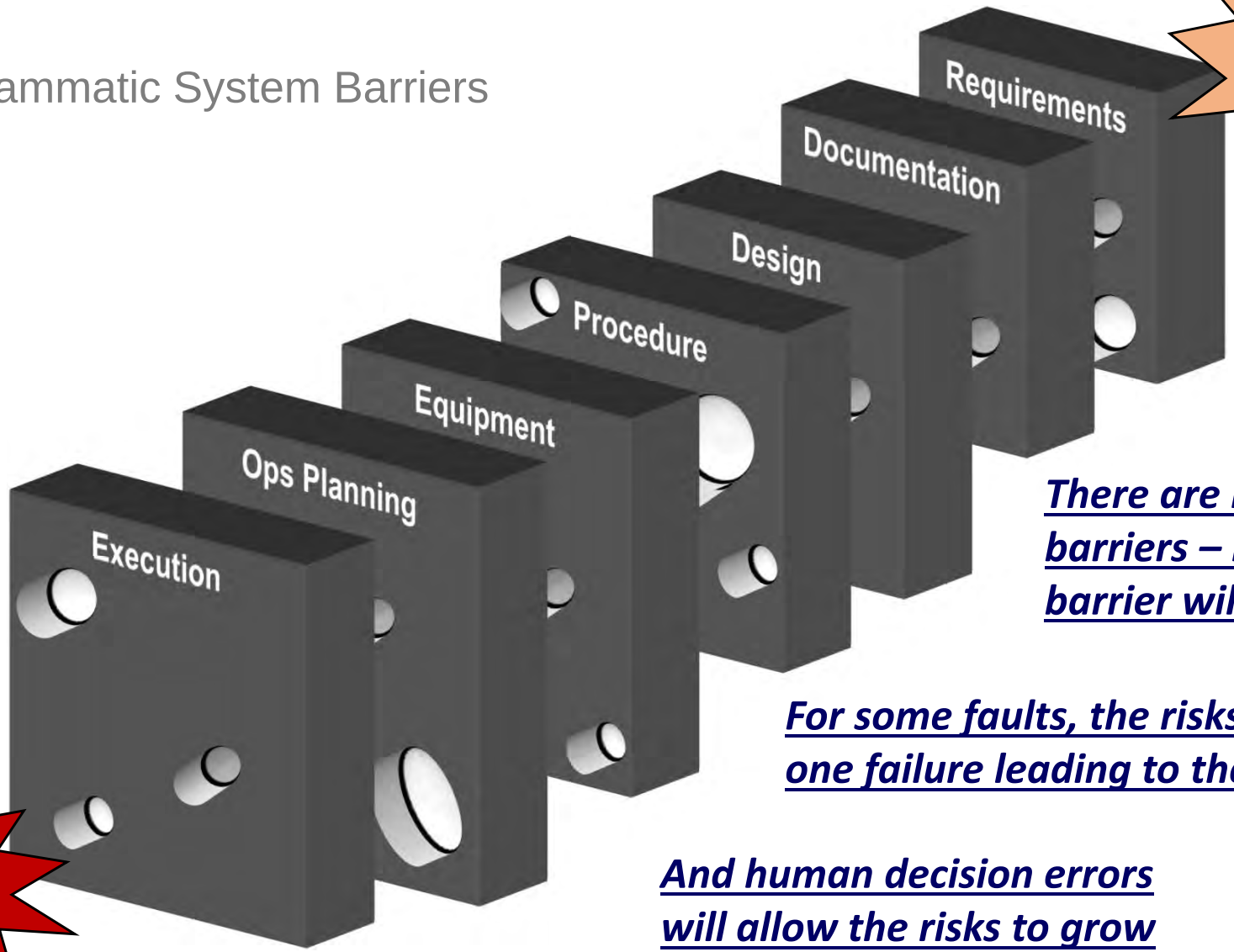
Airline Accident Rates (1960-2016)



Driving risk down is founded in "tuned" technologies, systems, ***and people***, regardless of the domain

“Barriers” to Block Failures

Programmatic System Barriers



System Flaws

There are holes in our barriers – risks that the barrier will fail

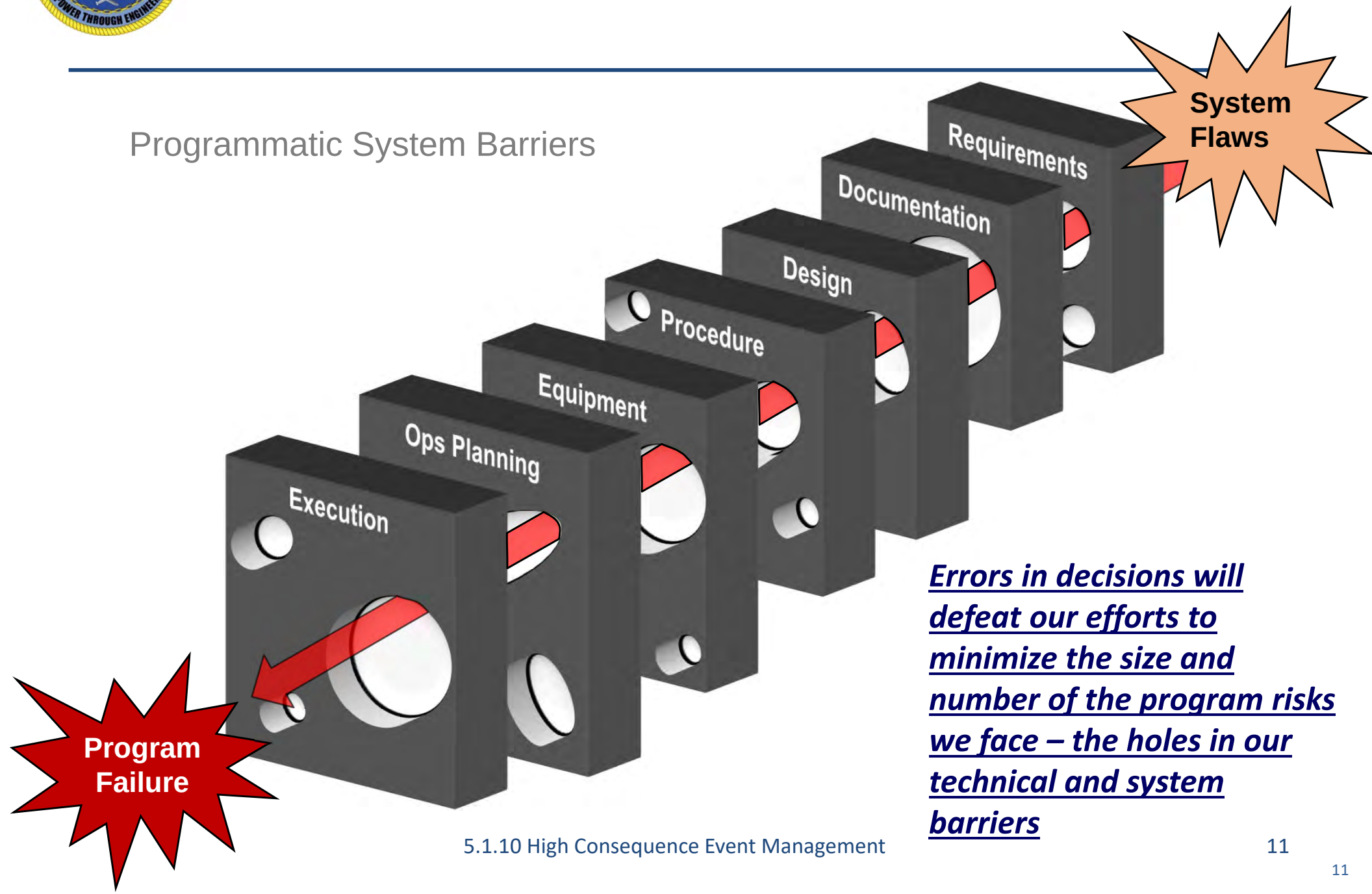
For some faults, the risks may align, one failure leading to the next…

And human decision errors will allow the risks to grow

Program Failure

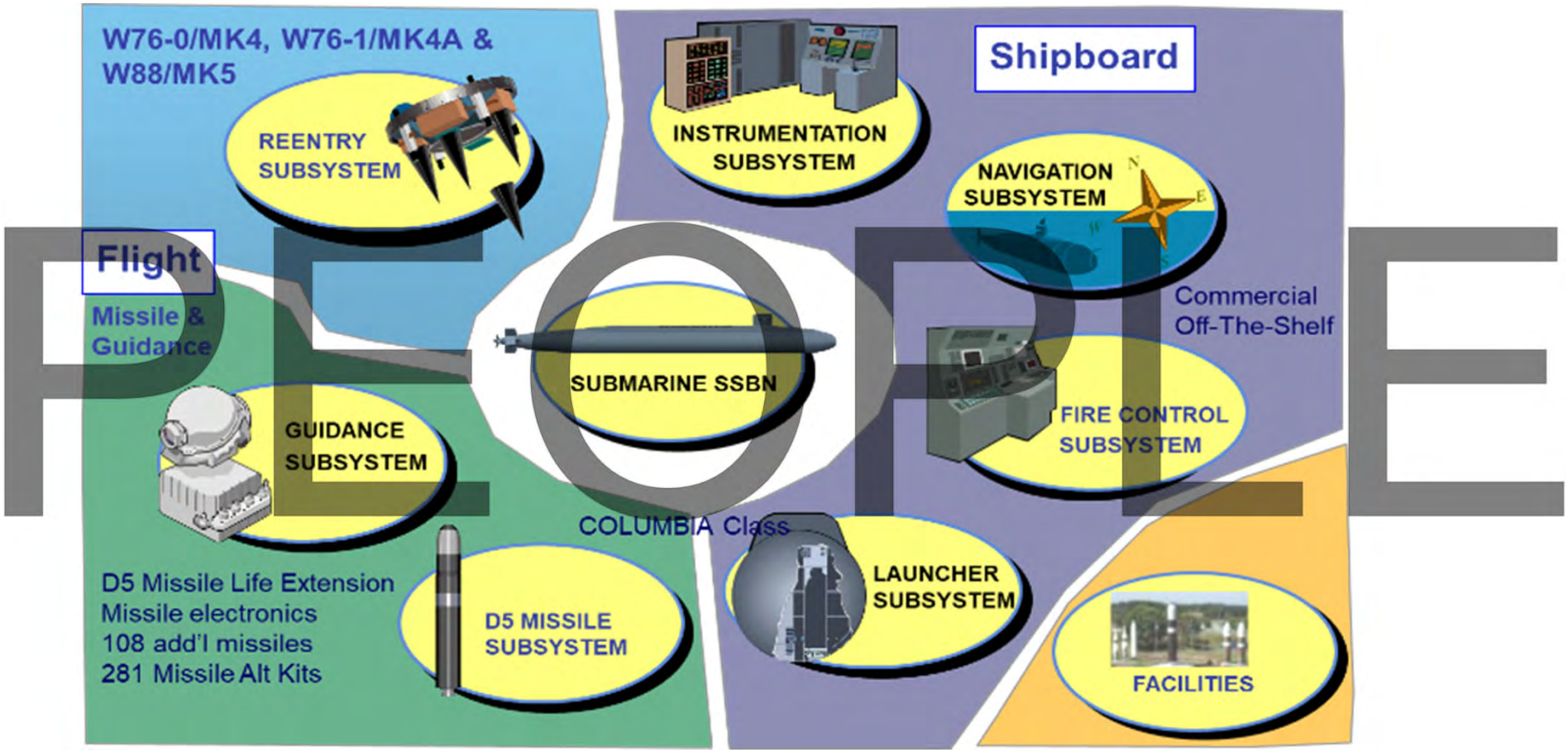
“Barriers” to Block Failures (cont.)

Programmatic System Barriers



Errors in decisions will defeat our efforts to minimize the size and number of the program risks we face – the holes in our technical and system barriers

Typical Navy System Elements



Culture must be thoughtfully developed, like every other example on this page.

Dimensions of Human Behavior



Risk Aware Behavior	VS	Human Element Weakness
LOOKING UP		
Questioning attitude		*Reflexive obedience*
Forceful back-up		*Reluctance to question authority*
Considered review of past decisions		*Sticking to past program decisions*
Transparent decision support		*Concealment of dissension*
LOOKING DOWN		
Encouragement of ideas and criticisms		*Insularity*
Openness to scrutiny and education		*Technical arrogance*
Invitation for benchmarking/innovation		*Not invented here*
Interrogation of the unexpected		*Success suffices*
Culture of risk evaluation		*Culture of production*
Integrated technical understanding		*Tribal knowledge*
Vertical knowledge and engagement		*Passive oversight*
LOOKING ACROSS		
Embrace of supportive, thoughtful process		*Surrender to bureaucratic process*
Formal, systematic risk engagement		*Informal or stove-piped treatment of risk*
Transparency and technical rigor		*Groupism*
Unambiguous execution of accountability		*Absence of accountability*
Output based evaluation		*Focus on inputs vice outputs*
Broad system ownership		*Not my problem*
Rigorous and open self-appraisal		*Disregard of honest appraisal*
LOOKING WITHIN		
Universal standards		*Situational values*
Loyalty to core values		*Misplaced loyalties*
Humility and leadership by example		*I'm above the rules*
Personal courage		*Who am I to judge?*
Public trust acceptance		*Others do it, must be OK*
Setting realistic, resourced goals		*Unreasonable demands*

How Subordinates Support Leaders

How Leaders Establish the Tone of the Organization

How Teams Organize and Operate for Effectiveness

How Individuals Guide Themselves (Ethics)

Applying Risk Aware Behaviors to Get Real, Get Better, and Build Learning Teams



High Consequence Events Prevention Framework	
Risk Aware Behaviors (vs)	**Human Element Weaknesses**
Get Real	
Align on Standards and Goals	
Integrated Technical Understanding	*Tribal Knowledge*
Public Trust Acceptance	*Others Do It, Must Be OK*
Setting Realistic, Resourced Goals	*Unreasonable Demands*
Culture of Risk Evaluation	*Culture of Production*
Find and Embrace the Red	
Personal Courage	*Who Am I to Judge?*
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Broad System Ownership	*Not My Problem*

Principles and Values



Principles of a Risk Aware Organization

Ownership
Leadership at all Levels

Empowerment
Responsibility, Authority, and Accountability

Mindfulness
Risk Aware Behaviors to Leverage Technical Strength

Assessment
Dynamic Risk Balance Between Safety & Production

Continuous Improvement
High Velocity Learning

As a Risk Aware Organization We Value…

Members who speak up, push back, and elevate risk issues if the approach is not right

Engaged supervisors who set the tone and standard for mindful behavior

Co-workers ready to identify and resolve unnecessary risk, even outside of their team

Individuals whose moral compass steers them, with integrity, to the right answer

HCEPF and Get Real, Get Better (GRGB)

GRGB

Get Real	Get Better	Build Learning Teams
Align on Standards and Goals	Use Proven Problem-Solving Methods	Encourage Learning through Trust and Respect
Find and Embrace the Red	Fix or Elevate Barriers	Specify Ownership



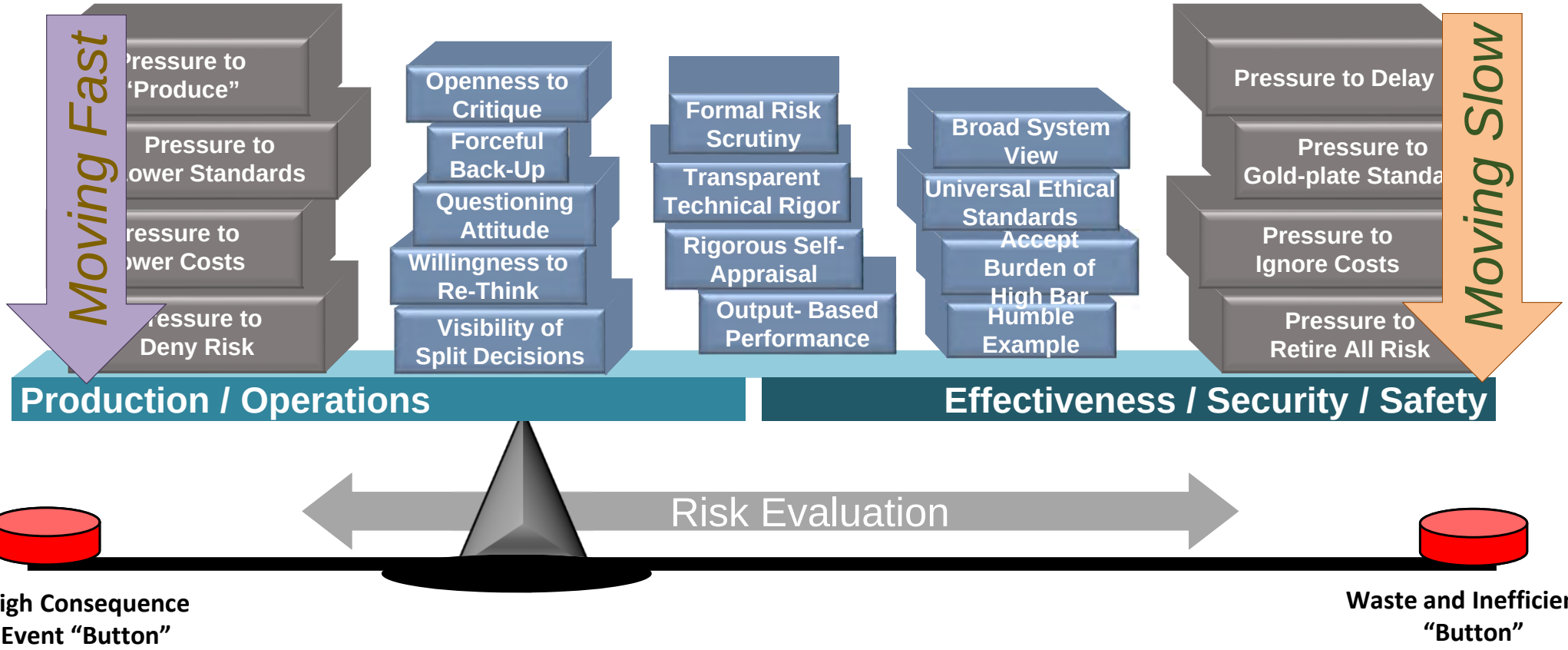
HCEPF

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- HCEPF is a cultural initiative priority of DIRSSP since implementation in 2012
- HCEPF has a structured Design with 24 Risk Aware Behaviors, each with a corollary Human Element Weakness
- The structure and measurement of HCEPF behaviors are how we track whether we are Getting Real, Getting Better, and Building Learning Teams
- HCEPF resources, including seminars, workshops, and tools can all be leveraged to foster the culture we need to ensure mission success
- HCEPF Risk Aware Behaviors are ***how*** SSP Gets Real, Gets Better, and Builds Learning Teams.

Balancing Differences in Risk Aversion



- Opposing forces are always present (and necessary)
- RABs help "stabilize" the balance
- Understand and manage the added risk of moving fast
- Moving fast may require thoughtfully adjusting the fulcrum

Key Finding

Need both Technical and Human Element Strengths

- Organizations reviewed had technical strength but did not deploy it well because of weaknesses in the human element
- An organization with both cross-functional technical skill and core human element strengths acts to:
  - ***Prevent the occurrence of high consequence events*** by avoiding the sequence of "closed switches" that enable such events
  - ***Respond effectively with resilience when faced with the unexpected***, coping with surprises for which there are no established procedures
- Root cause assessments typically identify technical failures but not the underlying Human Element causes that facilitated failures:
  - ***Technical/procedural focus****:* Technical errors, inattention, training gaps, and documentation shortfalls illuminated but not the underlying "why"
  - ***Human Element weaknesses left unaddressed****:* Shortstopping root cause assessment shortchanges utility, stymies change, and increases likelihood of recurrence

Mitigating risk of failure at an organizational, cultural level

OceanGate Submersible

OceanGate's submersible, Titan, was designed for depths up to 13,000 feet, specifically for transporting passengers to the Titanic wreck at 12,500 feet. Citing impediments to "innovation," the OceanGate CEO rejected adherence to proven commercial standards for the safe design, construction, and inspection of ships and deep-ocean submersibles. On 18 June 2023, on its third dive with five explorers onboard, including the CEO of OceanGate, Titan's unique carbon-fiber hull suffered a catastrophic implosion when approaching the Titanic, killing all onboard, including the CEO

What behaviors are present?



Kobe Bryant Helicopter Crash

On January 26, 2020, NBA star Kobe Bryant, his daughter, and six other passengers tragically perished in a helicopter crash while in route to a session at Bryant's basketball academy. Despite poor visibility due to fog, the pilot chose to attempt the flight, most likely motivated by his personal friendship and rapport with Bryant. The pilot, who was the company's chief pilot, had at least three opportunities to re-assess the riskiness of his plan. Rather than making an additional assessment of the risks, the pilot instead chose to continue with his previous decision to attempt the flight, navigating into a cloud, becoming disoriented, and crashing on a hillside

What behaviors are present?



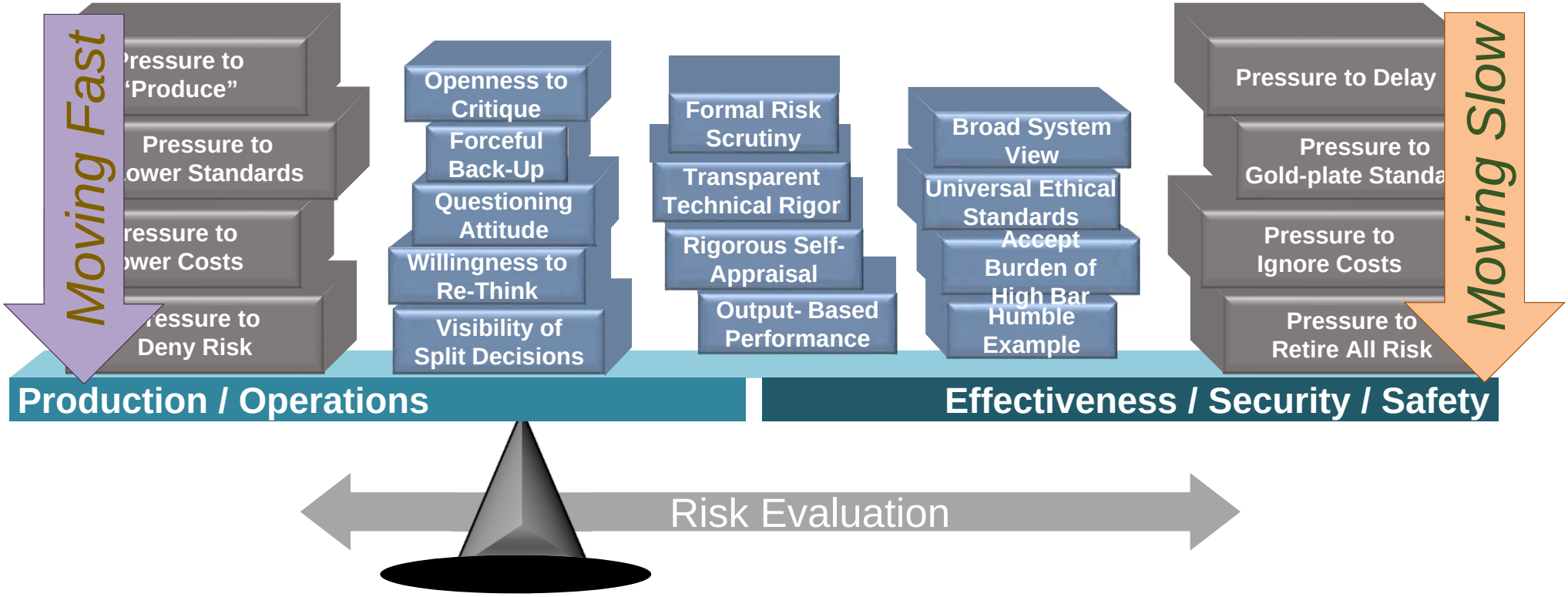
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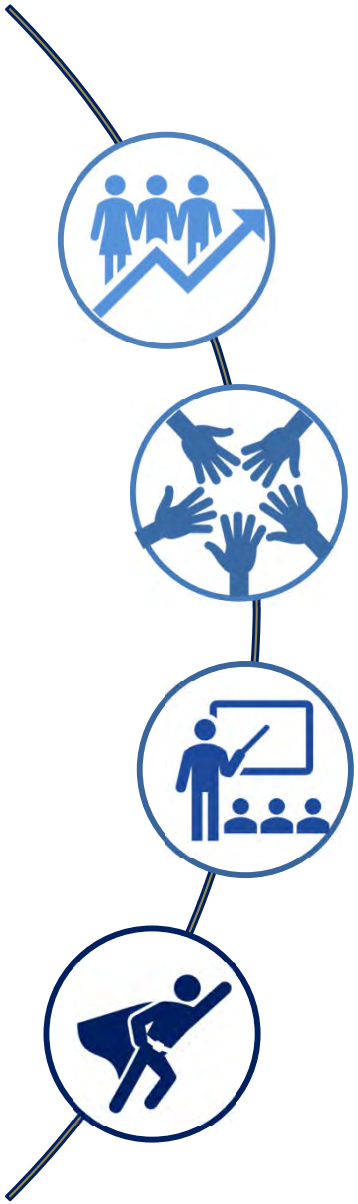
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Behaviors to Get Real

"Align on Standards and Goals"

Tribal Knowledge

Loss of B-2

Integrated Technical Understanding



Others do it, Must be OK

Y-12 Security Breach

Public Trust Acceptance



Culture of Production

Deepwater Horizon

Culture of Risk Evaluation



Unreasonable Demands

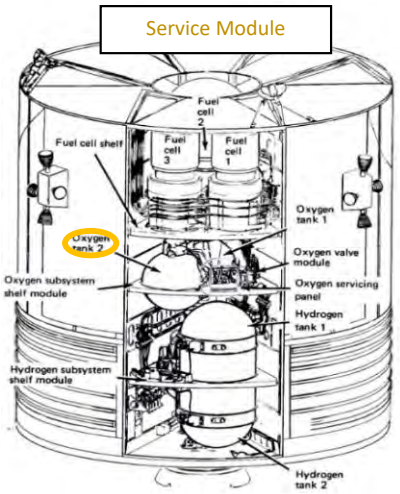
ICBM Crew Cheating

Setting Realistic, Resourced Goals

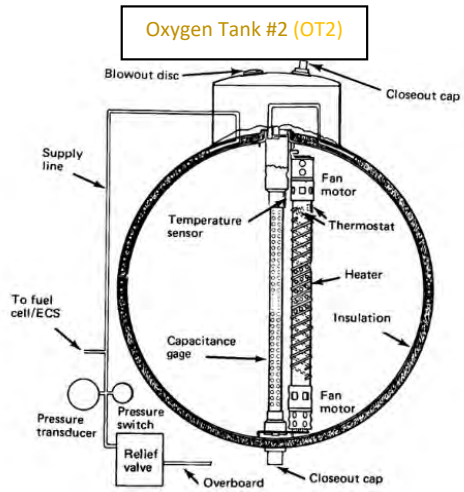


Apollo 13 Oxygen Tank Specification Error

In 1965, at the height of the Cold War and the Space Race with the USSR, NASA was moving fast and was far more accepting of risk. That same year, NASA raised the voltage requirement for the electrical components within the Apollo oxygen tanks from 28 to 65 VDC to speed testing; however, the contractor and NASA missed updating the specification for the internal thermostatic switch that cycled internal heaters at 80 ºF—the seeds of disaster were planted



Moving forward to 1970, during pre-launch testing at Kennedy Space Center and unknown to technicians, the thermostatic switch contacts welded shut from the higher voltage, scorching internal components (up to 1000 ºF). Then, 56 hours after launch, when an astronaut stirred the oxygen tank, a spark from damaged wires ignited an explosion—a mortal wound to the spacecraft. Miraculously, all three crew members survived



What behaviors are present?

Tools Reference Guide

- A reference for sixty ideas for tools
- Used to strengthen Risk Aware Behaviors
  - Individually, by teams, and organizationally
- Not intended to be a prescription
- Supports existing program processes
- Menu to mold and modify as needed

Tools

Individual	Function	Page
Act on Leemers		1
Bias Check		2
F.A.C.E.S.		3
Pause When Unsure		4
Personal Self-Assessment		5
Signature		6
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Individual and Team	**Function**	**Page**
Case Study Reviews		8
Decision Making		9
Job-Site Review		10
Mental Modeling		11
Procedure Review and Adherence		12
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Turnover Review		15
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Function Key: = Setting Expectations, = Assessment, = Feedback, = Coaching, = Broadening the View

Behaviors to Get Real

"Find and Embrace the Red"

Success Suffices

Loss of Shuttle Columbia

Interrogation of the Unexpected



Who am I to Judge?

Loss of Shuttle Challenger

Personal Courage



Situational Values

B-52 Air Show Rehearsal Crash

Universal Standards



Sticking to Past Decisions

Fukushima Reactor Accidents

Considered Review of Past Decisions



Misplaced Loyalties

Y-12 Security Breach

Loyalty to Core Values



Disregard of Honest Appraisal

Nuclear Logistics Failure

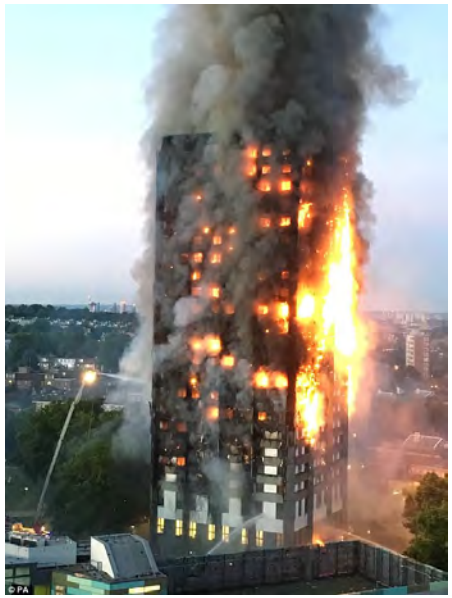
Rigorous and Open Self-Appraisal



Grenfell Tower Fire

The fire safety advice for Grenfell Tower residents was to “stay put” unless the fire was affecting their own flat. This advice was based on the assumption that every flat was designed as a “fire-resisting box” so that a fire ignited on one floor would burn out without affecting any other floors. This may have been true before external cladding was added during a refurbishment. Unfortunately, this cladding has been pointed to as a likely factor in how the flames spread up the front of the building at such a rapid pace

What behaviors are present?



Behaviors to Get Better

"Use Problem Solving Methods"

Reflexive Obedience

Inadvertent Weapons Move

Questioning Attitude



Focus on Inputs vice Outputs

Deepwater Horizon

Output-Based Evaluation



Informal or Stove-Piped Treatment of Risk

Drone Impact on Chancellorsville

Formal, Systematic Risk Engagement



Surrender to Bureaucratic Process

Y-12 Security Breach

Embrace of Supportive, Thoughtful Process



Y-12 Security Breach

Approved waivers allowed guards to physically monitor areas covered by inoperable cameras. With over sixty cameras out of commission, these waivers were still routinely approved despite the impracticality of the required level of physical monitoring with the existing guard force

What behaviors are present?



Behaviors to Get Better

"Fix or Elevate Barriers"

Insularity

Target Drone Impact on Chancellorsville

Encouragement of Ideas and Criticisms



Not Invented Here

Highly Enriched Uranium Facility Security Breach at Y-12

Invitation for Benchmarking/Innovation



Concealment of Dissension

Loss of Shuttle Challenger

Transparent Decision Support



Target Drone Impact on *USS CHANCELLORSVILLE*

Weapons test range controllers and briefers did not accurately communicate the risks presented by the exercise or disclose their ongoing problems with drone control, especially the signal delays, considering such issues to be internal matters of no concern to the ship

What behaviors are present?



Behaviors to Build Learning Teams

“Encourage Learning Through Trust and Respect”

I’m Above the Rules

Greeneville Collision

Humility and Leadership by Example



Technical Arrogance

Explosion and Sinking of Deepwater Horizon

Openness to Scrutiny and Education



Reluctance to Question Authority

Greeneville Collision

Forceful Back-Up



Behaviors to Build Learning Teams (cont.)

"Specify Ownership"

Groupism

Fukushima Reactor Accidents

Transparency and Technical Rigor



Not My Problem

Hurricane Katrina

Broad System Ownership



Absence of Accountability

B-52 Air Show Rehearsal Crash

Unambiguous Execution of Accountability



Passive Oversight

Hurricane Katrina

Vertical Knowledge and Engagement



GSA Contracting Failure

GSA was expected to pay Woodies Holdings, LLC for renting out 48,500 sq. ft. of office space to government agencies. Rather than calculating the payment by using the dictated “available office space,” a contract lease negotiator working on behalf of GSA instead used “total rentable space” to determine the amount owed. “Trusting” the contractor’s work, no one from GSA checked the numbers on any of the contracts, resulting in Woodies receiving millions of dollars more than they were meant to. Woodies was overpaid for 11 years until a GSA lease specialist noticed the mistake

***What behaviors are present*?**

***What tools can help to promote the corresponding Risk Aware Behavior*?**



Breakout Groups

Wells Fargo

In 2016 Wells Fargo Bank was implicated for fraud on a massive scale. Under the leadership of CEO John Stumpf the company had been making a push to sell more fee-bearing products, such as lines of credit or checking accounts, to their customers since 2007. From a public perspective this initiative appeared to be a success, with Wells Fargo boasting sales figures at four times the national average. However, high sales quotas for individual employees were enforced to such an unreasonable extent that employees at Wells Fargo branches, seeing no other way to meet the sales goals established by regional and district managers, began to open accounts and create new credit cards for existing customers without their authorization. From 2007 until 2016, Wells Fargo continued these unethical practices, until they were fined more than $2.8 billion in criminal and civil lawsuits

What behaviors are present?



Amazon Web Services

While conducting maintenance to increase the speed of Amazon Web Services' (AWS) new S3 server billing system, an engineer typed a command which unintentionally also resulted in the loss of two other S3 subsystems. With no fail safe in place, AWS lost its ability to support many of its customers when the shutdown affected a large portion of internet service, major websites, and smart-home appliances

AWS spent four hours correcting the error which resulted in an over $150M direct economic impact to S&P 500 companies and another $160M loss to US financial services companies. AWS immediately blamed the outage error on one employee instead of looking into how their entire (fragile) system could crash from a single mistake

What behaviors are present?

AAV Sinking During Exercise

In July of 2020, 13 amphibious assault vehicles (AAV's) launched from the amphibious transport dock ship, the USS Somerset, to San Clemente Island off the cost of San Diego, CA for a training exercise. After a long day of operation, the decision was made to leave 4 of the AAV's that were experiencing mechanical issues on the island, while the remaining 9 vehicles returned to the ship with no safety vessel present. On that return trip, one AAV's transmission seized due to a lack of transmission oil and began to take on water. After leaders waited nearly 30 minutes to order an evacuation, another AAV collided with the mishap AAV. Finally, a sea swell filled the mishap AAV with water, causing it to quickly sink along with eight Marines and a Navy Corpsman still wearing their heavy combat gear. The loss of life was completely avoidable

What behaviors are present?



Hurricane Katrina - Evacuation

Although Federal, State, and local officials were well aware that a large population, including many elderly and special needs residents, would fail to evacuate New Orleans ahead of a major storm, and that a major hurricane would flood the city, none of these levels of government planned for evacuating the flooded city. One (or all) of the concerned organizations should have taken responsibility for driving such planning

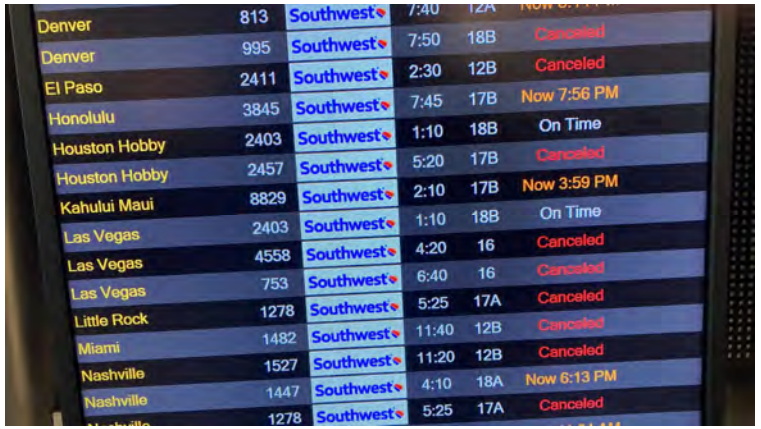
What behaviors are present?



Southwest Airlines Services Meltdown

During the 2022 holiday season, a punishing winter storm caused widespread flight cancellations across the United States. No airline suffered worse than Southwest, whose outdated computer system buckled under the immense number of cancellations and changes. Ultimately, Southwest lost nearly $825M due to the failure and millions of passengers were left alone to find a way to their final destinations. After the incident, it was clear that Southwest had prioritized their profit margins over investing in a more robust computer system, as theirs was clearly unable to handle the demands of modern flight scheduling. This showed that Southwest leaders were prioritizing profitability over their commitment to a positive customer experience

What behaviors are present?



Missouri Duck Boat Accident

On July 19, 2018, 17 tourists drowned when a duck boat (a WWII era vehicle designed for both land and water operation) capsized in a Missouri lake. While stormy weather and high winds that day were the natural causes, behaviors and attitudes of the duck boat owners also contributed to the tragedy. For instance, flaws in the duck boat's design were made obvious to the owners well before this incident. A private inspector had labeled the exhaust pipe as a sinking risk in high weather and an investigation from the National Transportation and Safety Board had determined that their canopy design would prevent people from escaping in the event of sinking. Despite these clear warnings, the duck boat owners prioritized successful business operations over customer safety. This also showed that, while operating the vehicle in its current condition was not illegal, the owners were willing to follow minimum requirements rather than holding themselves accountable for the potential loss of life

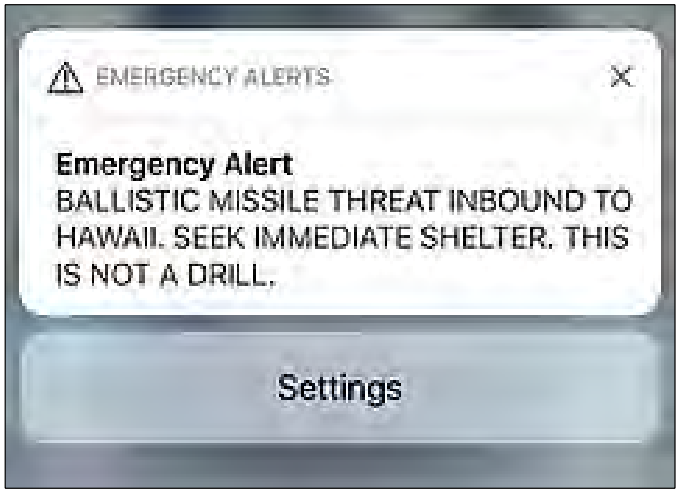


What behaviors are present?

Hawaii Missile Threat

In 2018, during a regularly scheduled test of Hawaii's Emergency Missile Warning System, a "real-world" ballistic missile emergency alert was sent to the public after a Hawaii Emergency Management Agency official selected the "Missile alert" option instead of the "Test missile alert" option. The message to the public read, "BALLISTIC MISSILE THREAT INBOUND TO HAWAII. SEEK IMMEDIATE SHELTER. THIS IS NOT A DRILL." A similar warning was also broadcast across the state's television screens. Due to poor coordination with FEMA, and the lack of an internal process to retract an errant message, it took 38 minutes to correct the mistake

What behaviors are present?



Merrimack Valley Natural Gas Explosion

In 2018, a Columbia Gas (CG) project to replace the old cast iron distribution main with new plastic piping didn't include reconnecting the sensing lines to control pressure in the low-pressure side of the distribution system. Without the sensing lines, pressure regulating valves sent high-pressure gas to houses and businesses that ruptured their gas pipes, damaging 131 structures, destroying five homes, and caused 22 casualties, including one fatality, in massive explosions





The investigation determined "weak engineering management" did not "adequately plan, review, sequence, and oversee the construction project." Specifically, the engineers and craft failed to ensure an important safety feature to regulate the gas pressure to the end users was restored. Basic system testing should have revealed the problem before catastrophe

What behaviors are present?

Air Force Failure to Investigate

In 2007, a B-52 bomber transported 12 Advanced Cruise Missiles to another base for maintenance. Operators thought none had nuclear warheads, but due to a series of errors, such as intermingling missile types in storage, half of them were nuclear armed—a surety violation. To make matters worse, shortly afterwards, a classified Minuteman III missile component was shipped to Taiwan in error. The rushed Air Force investigations into these incidents lacked depth in their analyses of root causes and left many convinced "that the primary motivation within the Air Force was to finish the reviews as quickly as possible, with as little further embarrassment as possible, and move on." Ultimately, the flawed investigations led to the forced resignations of the Air Force Secretary and Chief of Staff

What behaviors are present?



Notre Dame Cathedral Fire

A fire broke out at Notre Dame de Paris during a comprehensive renovation project. The only person monitoring the alarm panel, a security employee that had started three days prior, was confused by the indications and sent an on-site security guard to visually confirm the warning. Seeing no signs of smoke, the security guard returned, and the security employee contacted their supervisor for guidance. By the time the location of the fire was discovered, it was too late, and the resulting damage could require 20 years and $1.1 billion to restore. Of note, a past decision to omit sprinkler installation to the cathedral attic to preserve the aesthetics was not revisited while planning protective measures for the renovation work, a time when the risk of fire is elevated

What behaviors are present?



Missed Payroll

An employee of Luzerne County, Pennsylvania who worked in payroll took vacation on Black Friday after Thanksgiving. However, no one else was assigned the responsibility for payroll when this employee was not in the office. As a result, none of the 1,200 county workers received their paychecks on time, causing a host of problems. The head of finance assumed the employee would process the payroll while on vacation

What behaviors are present?



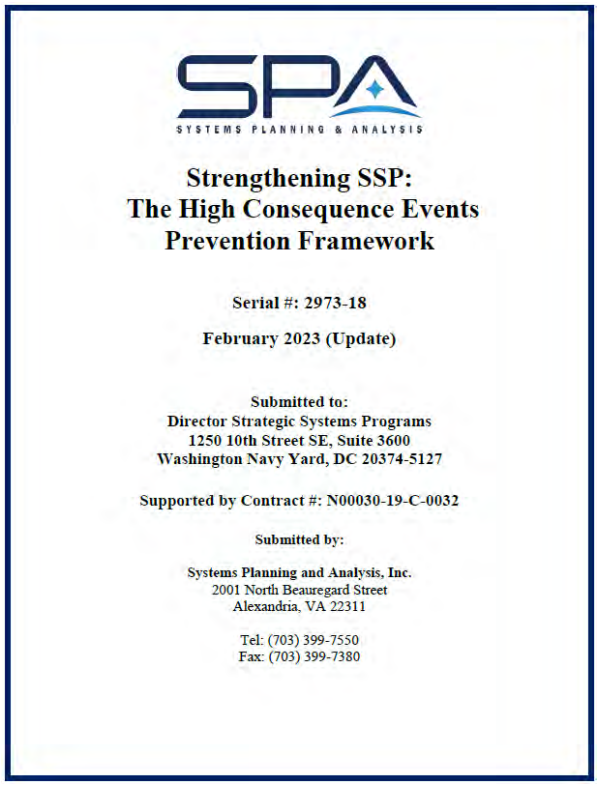
Osprey “Ghost 31” Crash- Norway 18 March 2022

A Marine MV-22B Osprey, call sign “Ghost 31”, crashed into the steep side of the Gråtådalen Valley in Norway during a training mission. Deviating from the flight plan, the uncertified crew decided to descend below altitude minimums and conduct (unauthorized) low-altitude tactics (LAT), filming their exploits with a GoPro camera—also unauthorized. Overcorrecting from a steep turn, the pilots put the Osprey nearly vertical (over 88 degrees), lost speed and altitude, and impacted the valley wall. The mishap resulted in the total loss of the $84M aircraft and the deaths of four Marines, all of which was captured on video recovered at the crash site a day later, after weather cleared

What behaviors are present?



Resources


SYSTEMS PLANNING & ANALYSIS

Strengthening SSP:
The High Consequence Events Prevention Framework

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Director Strategic Systems Programs
1250 10th Street SE, Suite 3600
Washington Navy Yard, DC 20374-5127

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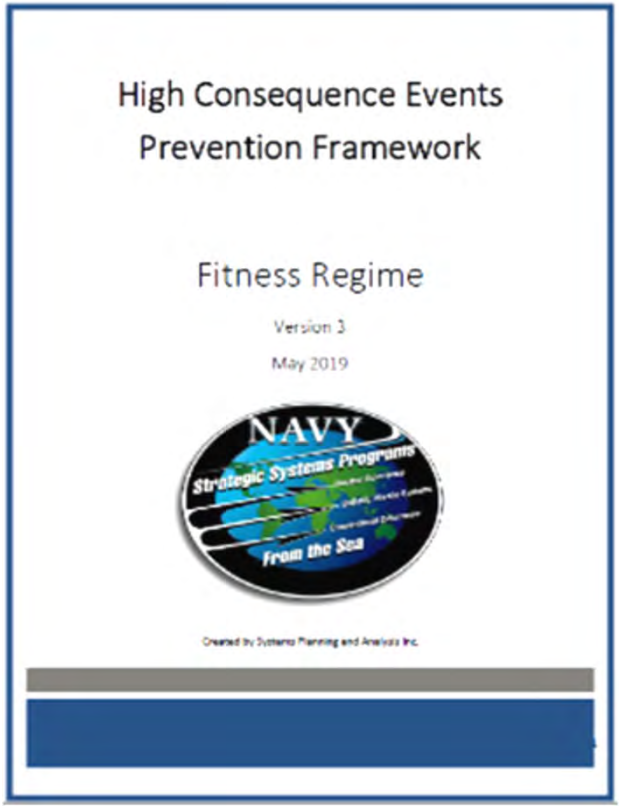
Submitted by:

Systems Planning and Analysis, Inc.
2001 North Beauregard Street
Alexandria, VA 22311

Tel: (703) 399-7550
Fax: (703) 399-7380

HCEPF Report
Framework details and case studies

Fitness Regime
Day-to-Day Implementation

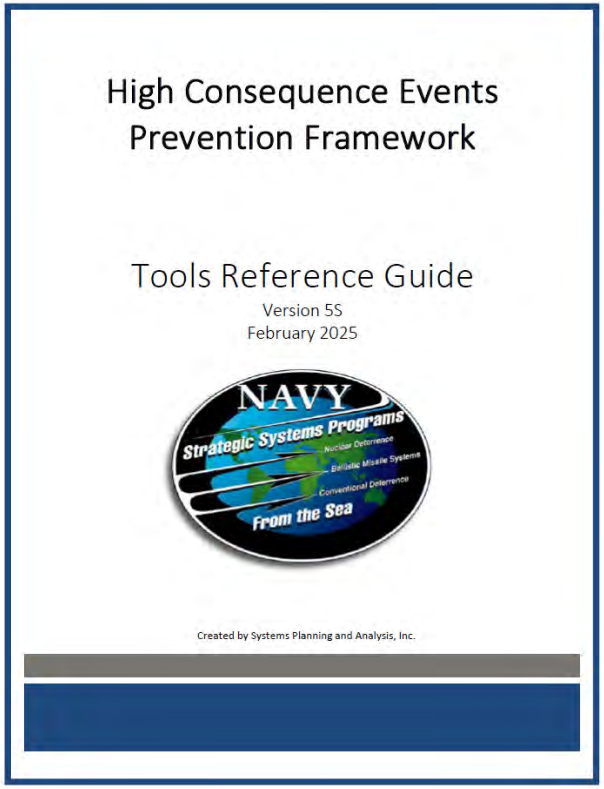

High Consequence Events Prevention Framework

Fitness Regime

Version 3

May 2019

NAVY Strategic Systems Programs From the Sea


High Consequence Events Prevention Framework

Tools Reference Guide

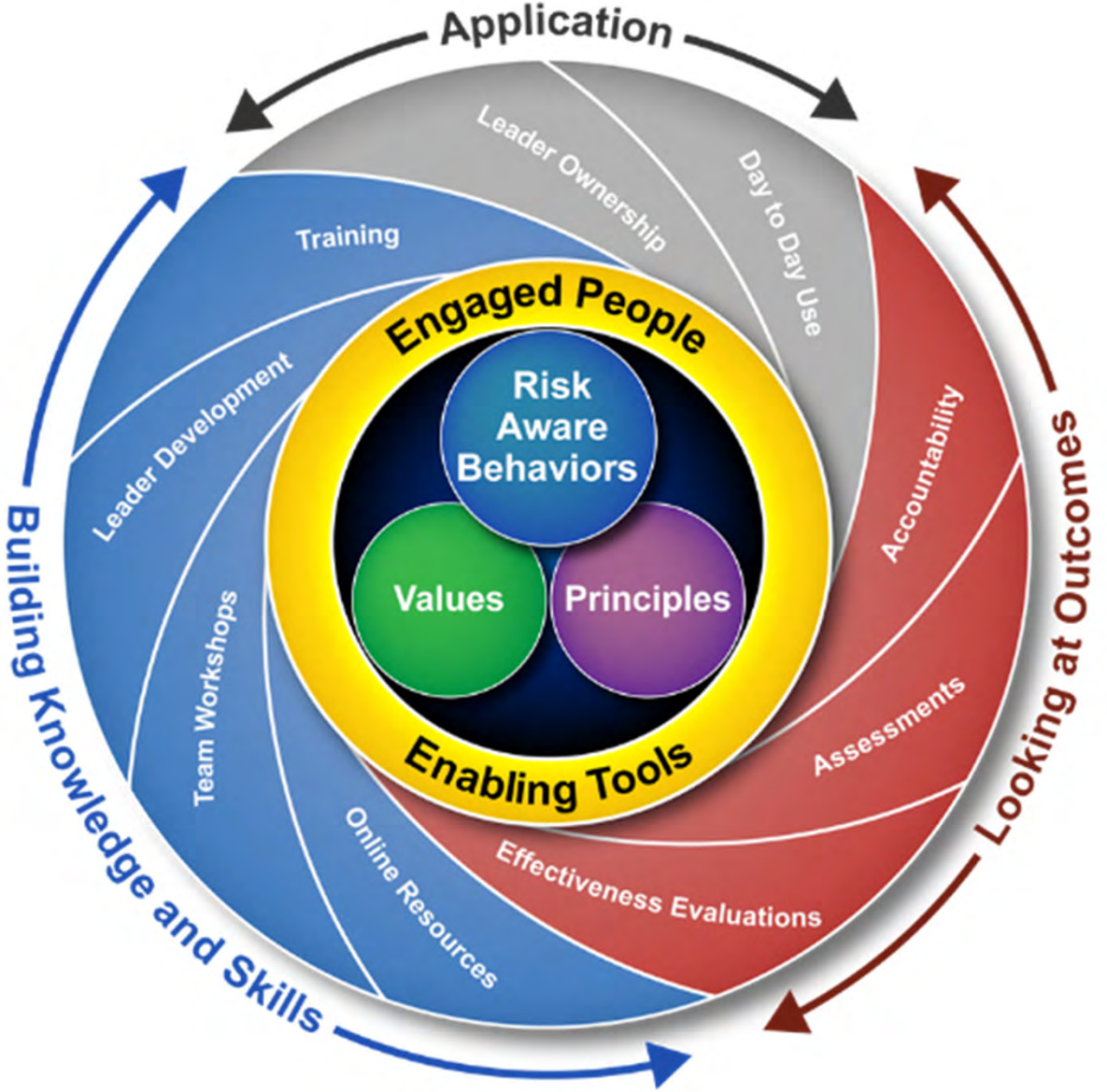
Version 5S
February 2025

NAVY Strategic Systems Programs From the Sea

Created by Systems Planning and Analysis, Inc.

Tools Reference Guide
60 tools for turning weaknesses into strengths

HCEPF Fitness Regime: Beating Entropy



- The deliberate design is a carefully crafted set of words and concepts
- Without our action the framework prevents nothing
- Our embrace of the design helps us achieve success by planning for peril

Four Keys for a Risk Aware Organization

- Senior leaders committed to and actively involved with HCEPF concepts
- Supervisors armed with the knowledge and skills to effectively lead and apply risk aware tools and aids
- Workforce convinced that events are preventable using skills to intervene and mitigate circumstances that are off course
- All hands empowered to recognize and mitigate risks and to intervene to ensure risks are not realized
  - At the technology, system, and people levels

Where the "rubber meets the road"

Back Up

Loss of Challenger and Columbia – Background

Challenger (1986) Summary:

Challenger was launched at air temperature of 36° F, 17° colder than any previous launch, despite evidence of previous O-ring burn and engineer concerns that the O-Rings in the Solid Rocket Booster (SRB) field joints between booster segments could fail. Following launch, both O-Rings failed in the aft joint on the starboard SRB, resulting in booster burn-through and a catastrophic explosion of the spacecraft

Columbia (2003) Summary:

A large piece of insulating foam detached from a support strut and struck Columbia during launch, damaging the leading edge of the left wing. Although the foam strike was reported following the launch, foam shedding was viewed as a routine occurrence that did not threaten the orbiter. Requests for high-resolution DoD imagery or other means to assess the damage were denied by NASA management. Heat-induced failure of the damaged wing during re-entry led to catastrophic breakup of the orbiter





Loss of Challenger and Columbia – Retro Report



Themes in the Shuttle Disasters

- The *Challenger* and *Columbia* disasters followed very similar causal trajectories
  - For both, technical teams were aware of increased risk of a specific class of failure that had been experienced before on several occasions
- After review, teams concluded that the risk was acceptable
  - In each case, the problem recurred more severely than past experience, and the decision to proceed resulted in a catastrophic accident
- The detailed investigation of the *Challenger* accident resulted in relevant reforms, which were gradually undermined over time
  - Underlying cultural and organizational weaknesses were present for *Challenger* and returned leading up to *Columbia*
- Recurrence reveals both the deep-rooted, cultural nature of Human Element weakness and the degree of leadership tenacity necessary for organizational change to stick

Loss of Challenger and Columbia Key Human Element Weaknesses

NASA events provided some 'textbook' examples:

- *Reflexive Obedience:* In both cases subordinate engineers did not question a "procedural output" ("acceptable risk") although it did not fit with their technical understanding of the problem or proper retirement of risk
- *Reluctance to Question Authority:* Courageous pushback might have saved lives. Many whose predisposition had been for greater safety emphasis were overridden by senior managers, and they did not push back
- *Sticking to Past Program Decisions:* The original shuttle design anticipated neither technical causal issue. When flights revealed the anomalies without flight failure, "normalization of deviation" meant subsequent incidents were viewed as evidence that the design was acting as predicted
- *Concealment of Dissension:* Competent engineers had recommended conservative courses of action that were omitted at Flight Readiness Reviews. Loss of insight as information moved up the hierarchy was 'regularized'

Loss of Challenger and Columbia Key Human Element Weaknesses (cont.)

- *Culture of Production:* Declaring the shuttle as an operational vice experimental system prior to Challenger was emblematic. While NASA leaders spoke to safety, personnel cutbacks sent other signals and the "faster, cheaper, better" NASA motto, at the time of Columbia, had an efficiency vice a risk focus
- *Insularity:* Both events revealed substantial evidence that differences in judgment on key issues were not just attributable to different individual technical understanding; instead fault lines existed between NASA organizations
- *Success Suffices:* Both Challenger and Columbia had looming patterns of failure that were disregarded as launches continued successfully. Eventually, the o-ring failure temperature dependence and the recurring external fuel tank foam separation problem (observed on nearly every flight), eventually caught up with the program on these missions
- *Surrender to Bureaucratic Process:* Proven engineering practices developed early in NASA's history were gradually translated to bureaucratic processes, and engineering judgment moved from the people to the process. Requiring proof that it was not safe to launch/reenter was emblematic

Loss of Challenger and Columbia Key Human Element Weaknesses (cont.)

- *Not My Problem:* Individuals concerned about risk unsuccessfully tried to move outside their "swim lane," reinforcing the general perception that assignment to a discrete role within the organization limited responsibility and accountability only within an individual's defined sphere
- *Who am I to Judge (Challenger):* Program managers presented a public face of relatively low risk for shuttle flights. Others in the organization tolerated this incorrect public disclosure despite their belief that actual flight risks were substantially higher
- *Technical Arrogance (Columbia):* Some leaders grew so confident in the excellence of their own engineering judgment that they developed a deaf ear to the safety "chorus" that came with the foam strike. They supported no real effort to understand the concerns raised or to evaluate whether or not existing analysis was adequate
- *Disregard of Honest Appraisal:* Appraisal processes were insufficient to institutionalize the learning that came from the Challenger catastrophe. Cultural realignment after Challenger did not stick

How could these weaknesses appear in your line of business?

Vignette Discussions

- Who, What, When, Where, Why, How?
- What kind of culture existed in the organization based on the reading?
- What behaviors led to this event?
- What corrective actions could have helped to mitigate the issues?
- WWRAOD – ***What Would a Risk Aware Organization Do?***